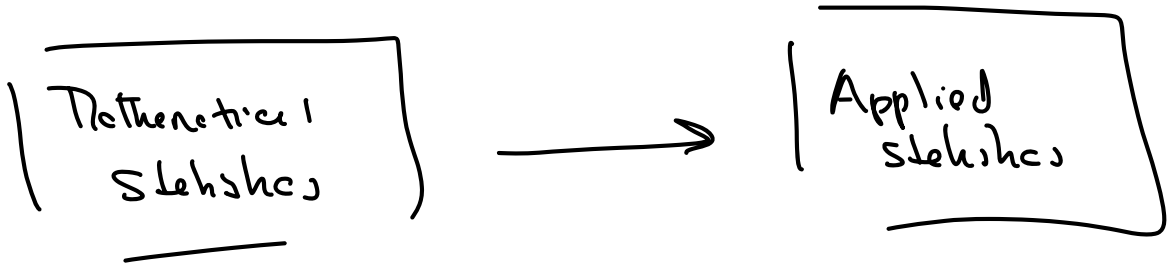
4 - Fundamentals of Mathematical Statistics

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Agenda



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Suppose you're an analyst hired to help improve cafeteria operations.

- Your focus: the **waiting times in line**.

1. **Point estimation**
   → What's the *average* wait time in the cafeteria queue?
2. **Choosing among estimators**
   → Which approach gives us the most *reliable* estimates?
3. **Confidence intervals**
   → How *sure* are we about those estimates?
4. **Hypothesis testing**
   → Have wait times improved between 2024 and 2025?

Recommended readings: Wooldridge Appendix C

1. Point Estimation

Estimating Queue Performance from Sample Data

You're helping the cafeteria team improve their queue management strategy.

We want to understand:

- What is the **average wait time** in the cafeteria line?
- How much do wait times **vary across customers**?
- How can we get **reliable estimates** without observing everyone?
- How does **sample size** affect our estimate?
- How can we avoid being misled by **random variation**?

To answer these questions, we turn to **point estimation**.

What Is Point Estimation?

Point estimation uses **sample data** to produce a single best guess — a **point estimate** — of an unknown population value (a **parameter**).

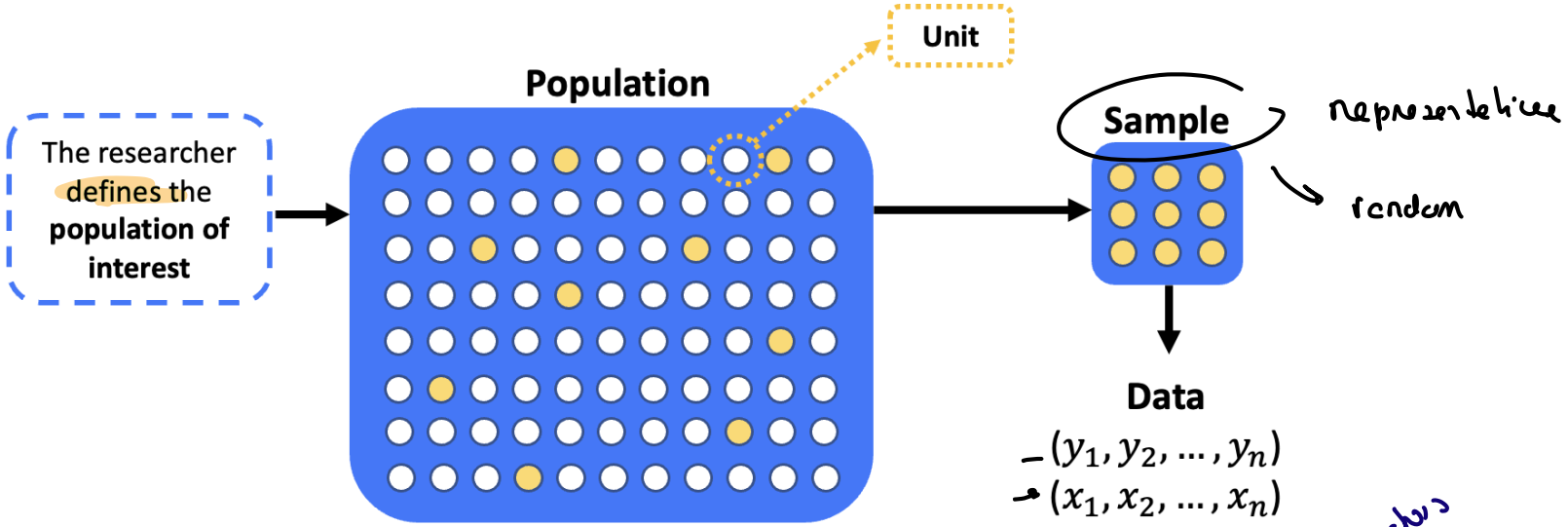
It's a core tool for **data-driven decisions** when we can't measure everyone.

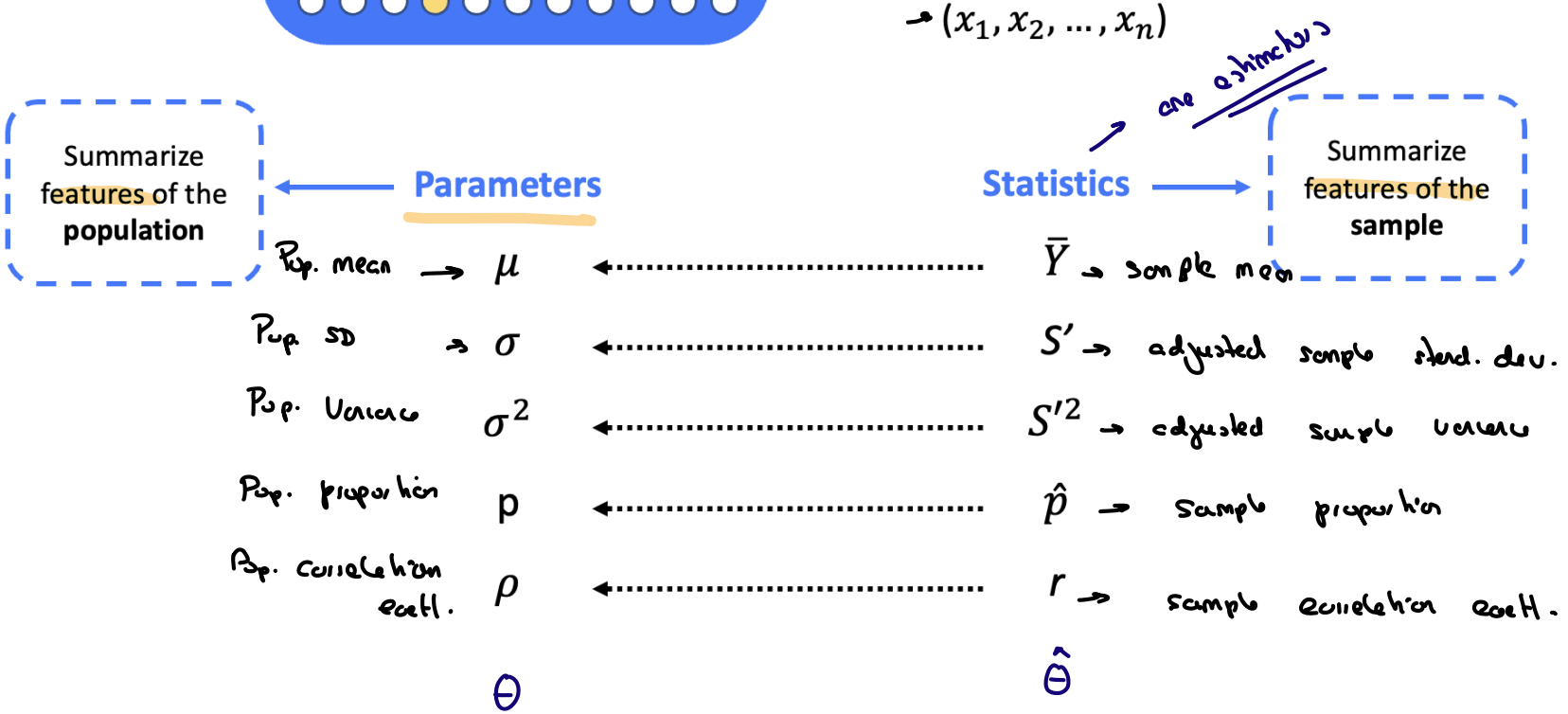
- **Example**: Estimating average wait times supports **staffing**, **scheduling**, and **process improvement** decisions.

What does point estimation involve?

1. Drawing a **sample** from the population
2. Choosing a suitable **estimator** (e.g., sample mean)
3. Using the data to compute the **estimate**
4. Evaluating how **reliable** it is (leads into confidence intervals)

Key Concepts

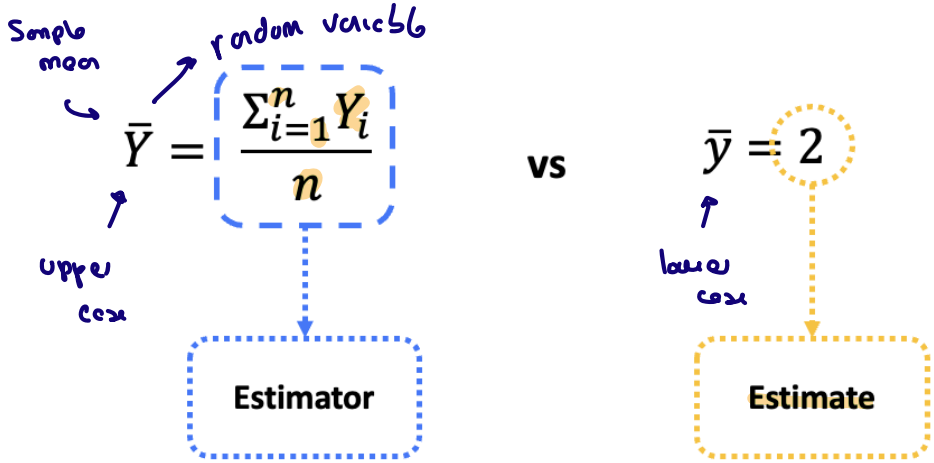




Estimator vs. Estimate

Point estimator: A rule or formula (like the sample mean) used to estimate a parameter

Point estimate: The specific number we get when applying the estimator to a sample



In our case:

- Let Y be students' **wait time in the cafeteria line** (in minutes).
- We use the sample mean $\bar{Y} = \frac{1}{n}\sum Y_i$ as the *estimator*.
- From the data, we calculate $\bar{y} = 6.8$ minutes $\rightarrow$ our *point estimate*.

What Is a Random Sample?

To estimate wait times reliably, we need a **random sample**.

Our **population** includes all university members (students, faculty, staff), and each individual is a **unit**.

We assume each unit's wait time comes from an **underlying distribution** — a **data-generating process** we can't observe directly.

A sample is **random** if:

- All units have the **same chance** of being selected
- Selections are **independent**
- Every sample of size n is **equally likely**

This allows us to treat $Y_1, Y_2, ..., Y_n$ as **independent, identically distributed (i.i.d.)** random variables.

Sample Statistics Are Random Variables

When we compute a statistic like the **sample mean**, we're using **just one sample** — but many samples are possible.

- The sample mean $\bar{Y}$ **varies from sample to sample**
- That's why $\bar{Y}$ is a **random variable**

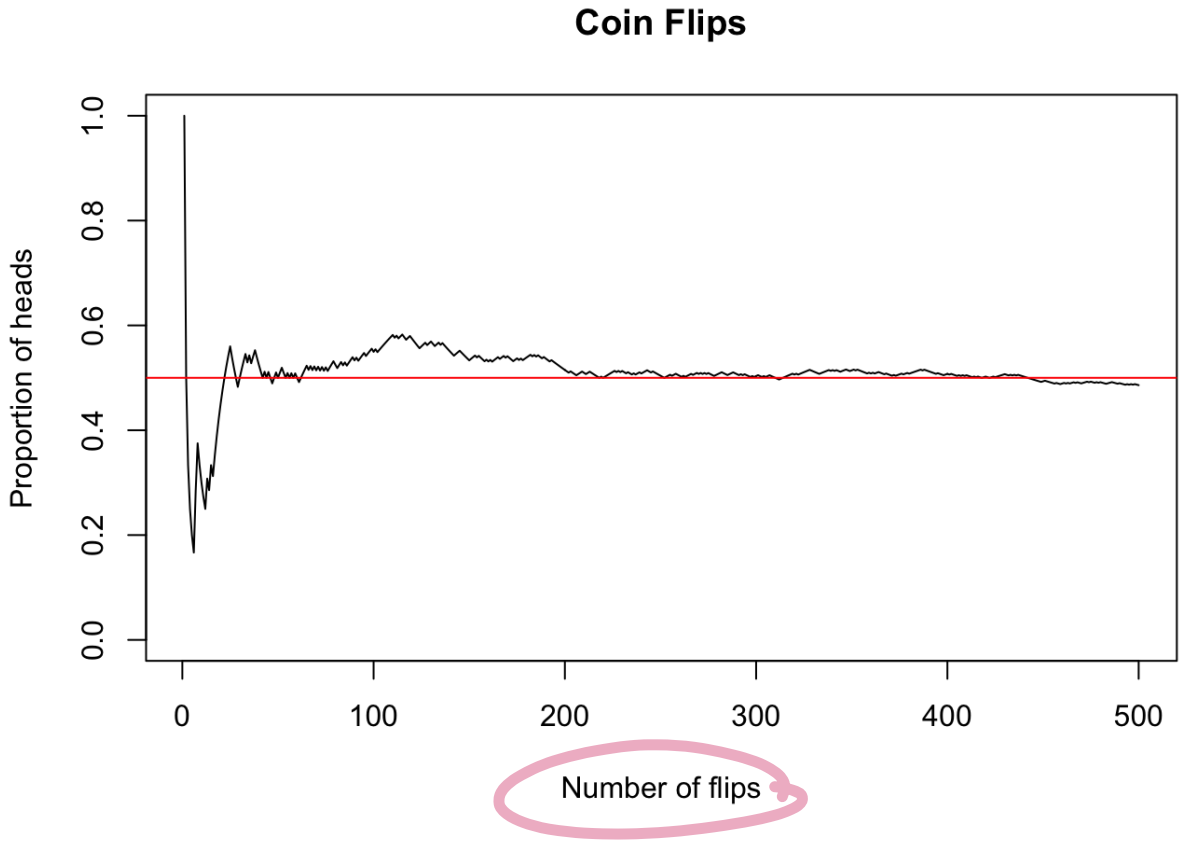
Sampling error = the difference between our estimate and the true population value $\rightarrow$ Caused by **random chance** in who ends up in our sample

- Our estimate is just **one realization** from a broader random process
- Sampling error is **not a mistake** — it's **unavoidable** when working with samples
- We **can't observe it directly**, since the true parameter is unknown
- But we *can* try to **reduce it** — and that leads to our next idea...

Law of Large Numbers

As sample size n increases, the sample mean, $\bar{Y}$, gets closer to the true mean, μ.

- Larger samples tend to reduce sampling error
- Our estimate becomes **more stable and accurate**



Sampling Distributions

A **sampling distribution** is the **probability distribution** of a statistic (like the **sample mean** $\bar{Y}$) across repeated random samples.

- It describes the **data generating process** for statistics like $\bar{Y}$
- It tells us how much our estimate would vary **across samples**
- It has its own **mean** (expected value) and **standard deviation** (called the **standard error**)
- It's the foundation for tools like **confidence intervals** and **hypothesis tests**

Try this **interactive simulator** to see sampling distributions in action.

Recap: Why Point Estimation Matters

You're helping the cafeteria improve its **queue management strategy** — but you can't ask everyone on campus.

To make data-driven decisions, you need **reliable estimates** from a **random sample**.

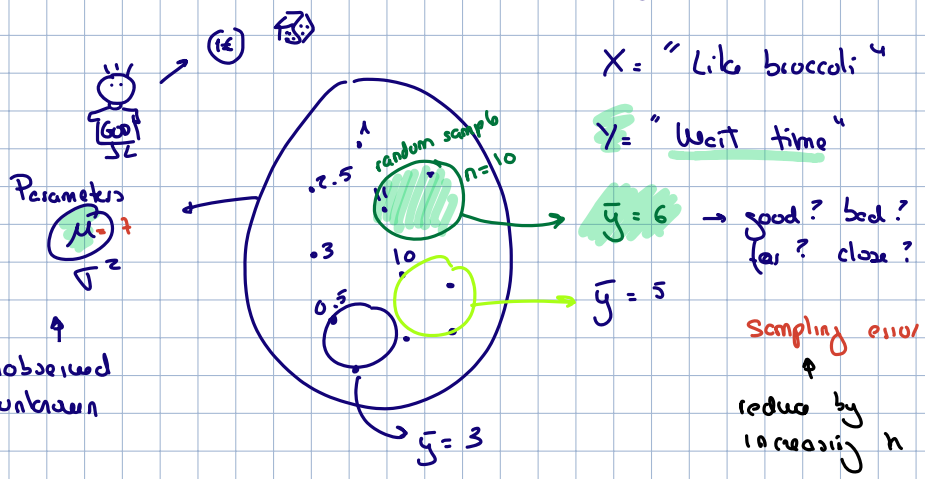
Point estimation gives you a toolkit to:

- Estimate the **average wait time** in the cafeteria line
- Understand **how much wait time varies**
- Think critically about **sampling error** and **variability**
- Appreciate the role of **sample size** in improving accuracy

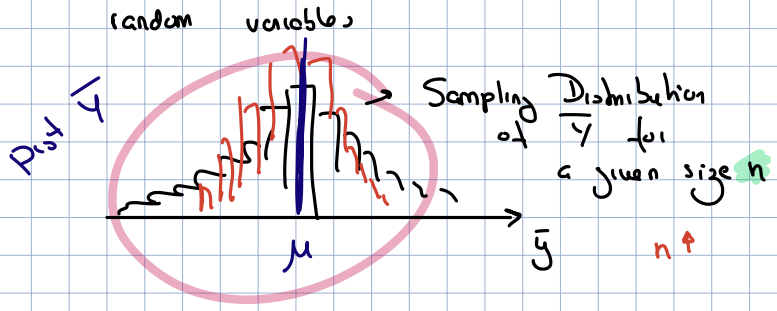
But... not all estimators are equally good. Before we build tools like confidence intervals or hypothesis tests, we need to ask:

What makes one estimator better than another?

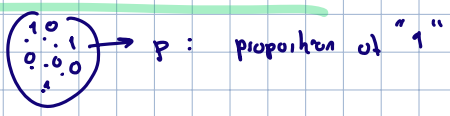
Fundamentals of Meth. Stchistics



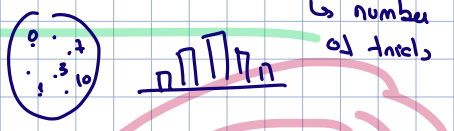
Estimators (like the sample mean, $\bar{Y}$) are random variables



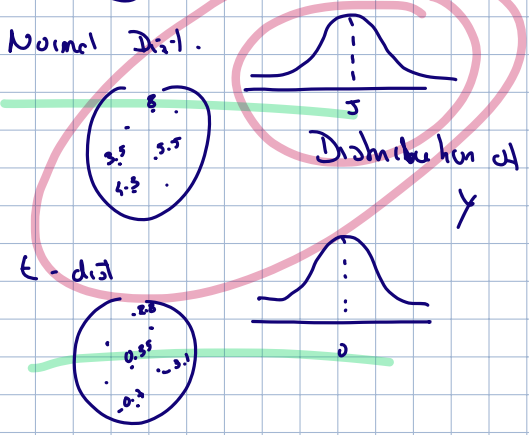
Bernoulli p



Binomial p, n



Normal Dist.



Uncertainty
↳ Reduce
↳ Quantify / measure
↳ Communicate

$\bar{Y} \pm$ ME → How should it behave?

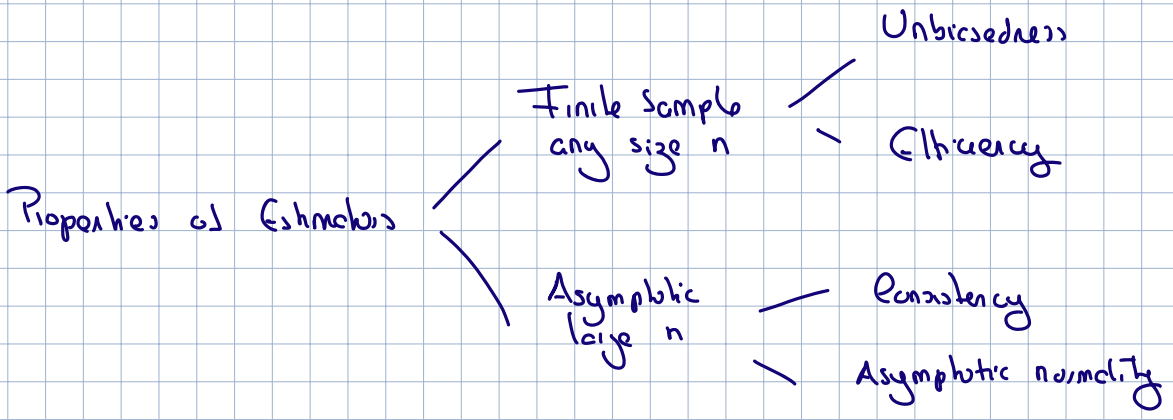
Uncertainty ↑ → ME ↑
Uncertainty ↓ → ME ↓

3 ingredients:
- sample size — n ↑ → ME ↓
- σ^2 — σ^2 ↑ → ME ↑
- confidence ↳ trade-off between precision and confidence

$$\bar{Y} \pm \underbrace{z_{\alpha/2} \text{ or } t_{\alpha/2} \times \frac{\sigma \text{ or } s}{\sqrt{n}}}_{ME}$$

90%
95%
99%

2. Choosing Among Estimators


Properties of Estimators
Finite sample any size n
Unbiasedness
Efficiency
Asymptotic large n
Consistency
Asymptotic normality

Not All Estimators Are Created Equal

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You're estimating the **average wait time in the cafeteria line** using sample data. You could:

- Take the **sample mean**
- Use the **median**
- Add weights by frequency of visits

But which estimator is **best**?

- Which one gives us the **most accurate**, **most stable** estimate?
- Can we trust it in **small samples**? In **large samples**?
- How do we compare alternatives?

To answer these, we need to study the **properties of estimators** like *unbiasedness*, *efficiency*, and *consistency*.

- They help us understand **how good** an estimator really is and why we often choose the sample mean.

Finite Sample Properties (1/3)

To choose a good estimator, we first ask: *what properties should hold, no matter the sample size?*

These are called **finite sample properties**, they help us compare estimators even when working with small or moderate samples.

Unbiasedness: on average, the estimator gets the right answer.

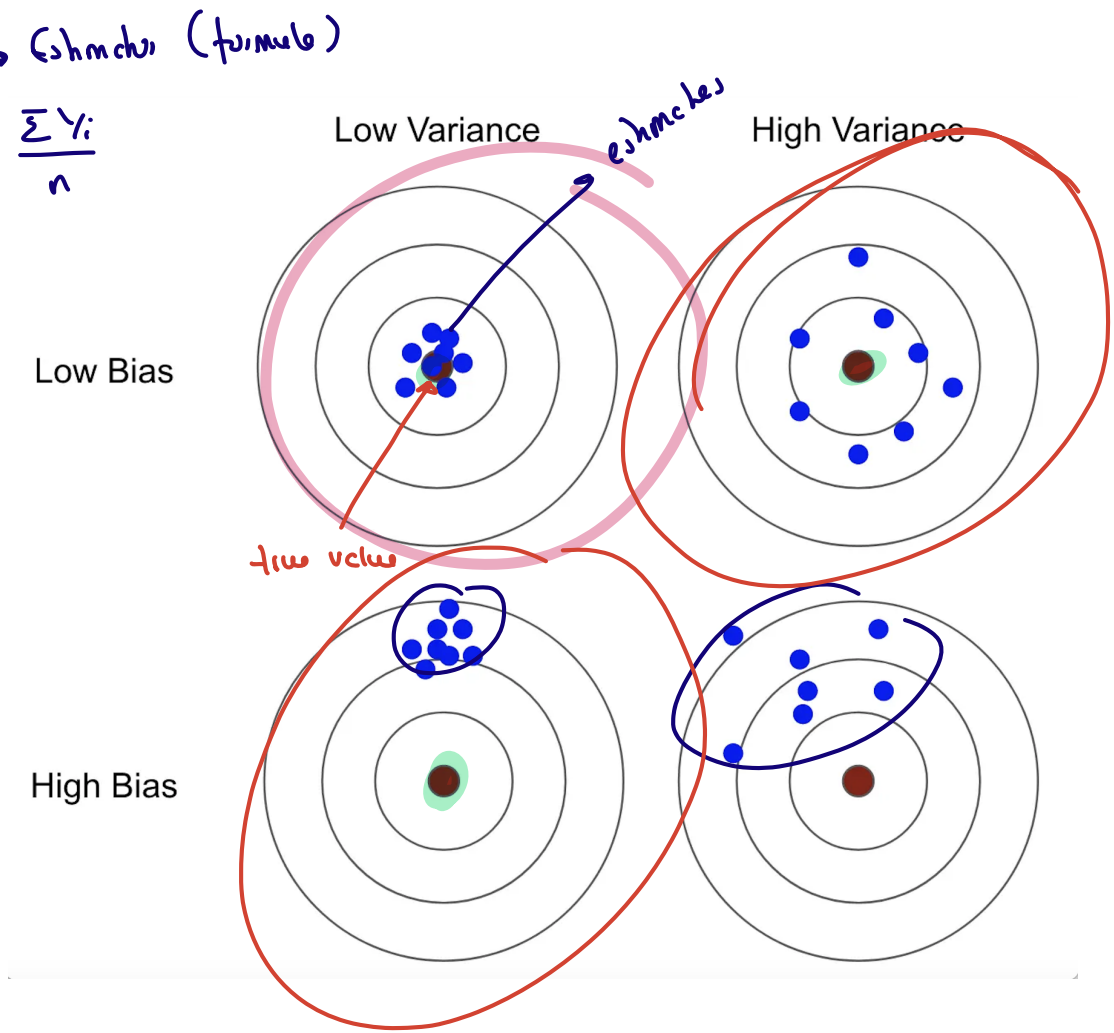
$$E[\hat{\theta}] = \theta$$

Efficiency: among unbiased estimators, we prefer the one with the **smallest variance**.

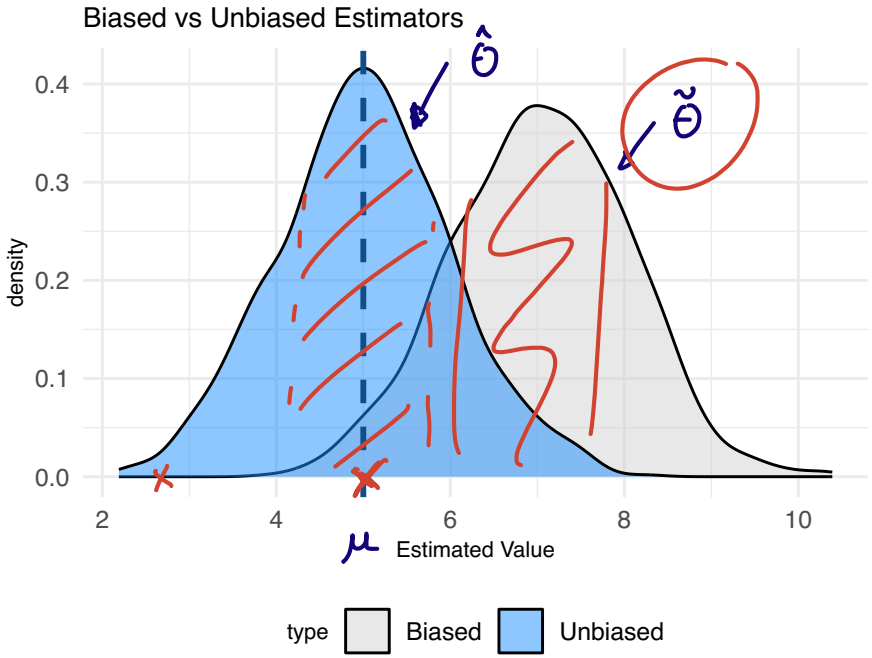
- When comparing among biased (or biased vs unbiased) estimators, we use the **mean squared error (MSE)**:

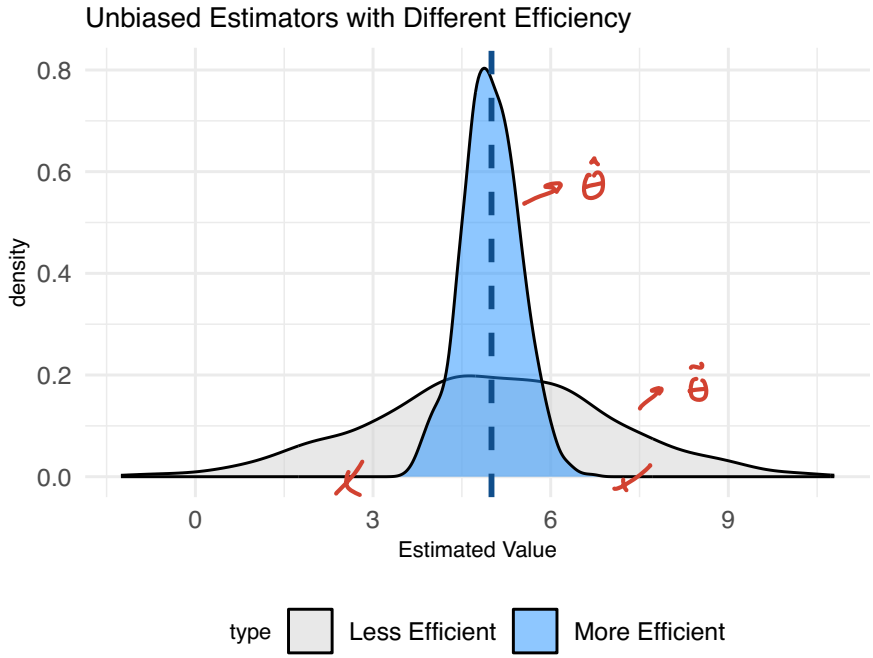
$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

Finite Sample Properties (2/3)



Finite Sample Properties (3/3)





- **Left**: note how the biased estimator is consistently off-target.
- **Right**: both estimators are unbiased, but one is more precise.

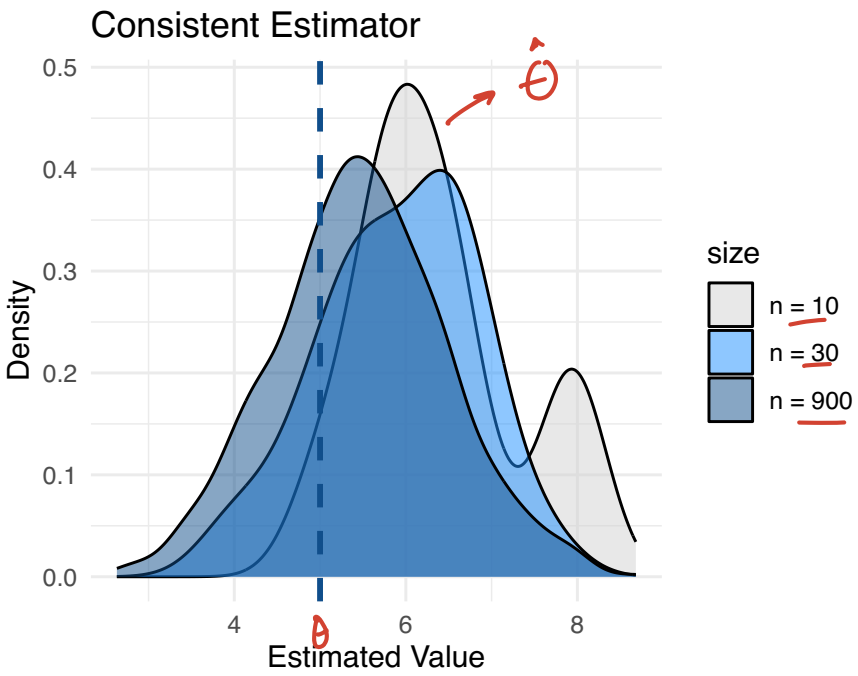
Asymptotic Properties (1/2)

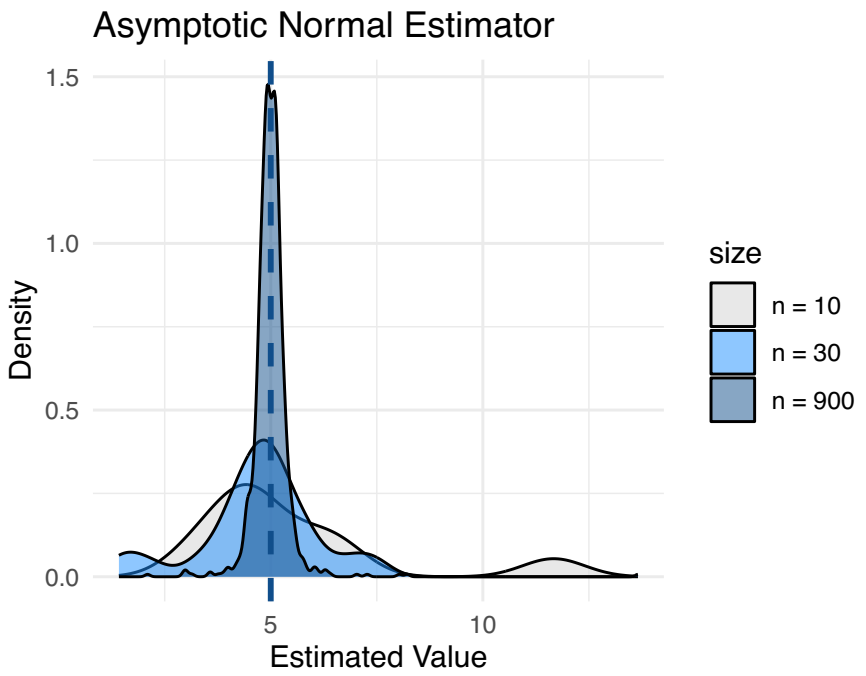
What happens to our estimators as the **sample size grows**?

These properties help us judge estimator behavior in **large samples**:

- **Consistency:** an estimator is **consistent** if it converges in probability to the true value as $n \to \infty$.
  - That is, with enough data, it gets arbitrarily close to the truth.
- **Asymptotic Normality** as n increases, the sampling distribution of the estimator becomes approximately **Normal**.
  - This lets us use inference proceedures based on the Normal distribution (e.g., t-tests, CIs) even when the original data is not Normal.

Asymptotic Properties (2/2)





- **Left**: a consistent estimator becomes centered on θ as n increases
- **Right**: the sampling distribution becomes approximately Normal as n grows.

Central Limit Theorem (CLT)

The **Central Limit Theorem** is the foundation of statistical inference. It tells us:

> *Even if the population distribution is unknown or skewed, the **sampling distribution** of the sample mean $\bar{Y}$ becomes approximately **Normal** as the sample size n increases.*

If $Y_1, Y_2, ..., Y_n$ are i.i.d. with mean μ and variance σ^2, then:

$$Z_n = \frac{\bar{Y}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

Or equivalently:

$$\bar{Y}_n \dot{\sim} N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{as } n \to \infty$$

Why the CLT Matters

- The CLT gives us a **known distribution** for $\bar{Y}$ — even when Y is skewed or unknown
- It's what allows us to build **confidence intervals** and perform **hypothesis tests**
- Without it, most of classical inference wouldn't work

In our case:

Even if the distribution of **wait times** is skewed, with a large enough sample the **average wait time in the cafeteria line** $(\bar{Y})$ is approximately Normal

$\rightarrow$ This lets us make **reliable staffing and scheduling decisions**.

Recap: Why Estimator Properties Matter

To make good decisions (like improving cafeteria wait times), our estimates should be:

- **Unbiased** $\rightarrow$ Not systematically off
- **Efficient** $\rightarrow$ Low variability across samples
- **Consistent** $\rightarrow$ Gets closer to the truth with more data
- **Asymptotically Normal** $\rightarrow$ We can rely on the Normal distribution for inferences

What's Next?

We've focused on **getting an estimate**. Now we ask:

- **How precise is it?**

Let's move to **confidence intervals**.

3. Confidence Intervals

Inference

Population (μ, σ^2)

↑ Sample $(\bar{Y}, S'^2, S')$: ≠ samples ≠ estimates

Are RANDOM VARIABLES

↳ Have a SAMPLING DISTRIBUTION

- $E[\bar{Y}] = \mu$
- $V(\bar{Y}) = \frac{\sigma^2}{n}$

↳ $SE(\bar{Y}) = \frac{\sigma}{\sqrt{n}}$ (σ: unknown; ↓ ↑)

CLT: $\bar{Y} \dot{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$ for sufficiently large n

↳ $\bar{Y} = \frac{\sum Y_i}{n}$

Sample mean of wait time

Y = "wait times"

CI : estimate ± margin of error

↳
- sample size ↑ → SE ↓ → ME ↓
- variability ↑ → SE ↑ → ME ↑
- confidence ↑ → critical value ↑ → ME ↑

CI for μ : $\bar{Y} \pm t_{\frac{\alpha}{2}} \frac{S'}{\sqrt{n}}$ ($\frac{S'}{\sqrt{n}} = \hat{SE}$)

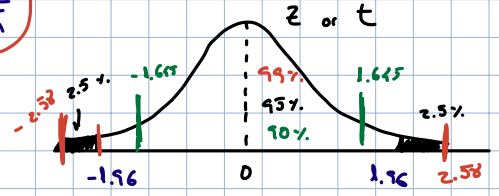
$\bar{Y}$: estimator of μ ; $t_{\frac{\alpha}{2}}$: critical value

99% : $z_{\alpha/2} = 2.58$

95% : $z_{\alpha/2} = 1.96$

90% : $z_{\alpha/2} = 1.645$

t_n ✓



From Point Estimates to Intervals

We observed students' **wait times in the cafeteria line** and found that the **sample average** wait time is **6.8 minutes**.

But, how confident can we be this reflects the **true population average wait time**?

- Sample means **vary** across samples
- Larger n improves accuracy, but doesn't remove uncertainty

So instead of asking:

- *What's the average wait time for students in line?*

We ask:

- *What range of values is the true average wait time likely to fall in?*

Let's use a **confidence interval** to quantify our uncertainty.

Real Data Example: Loading the Data

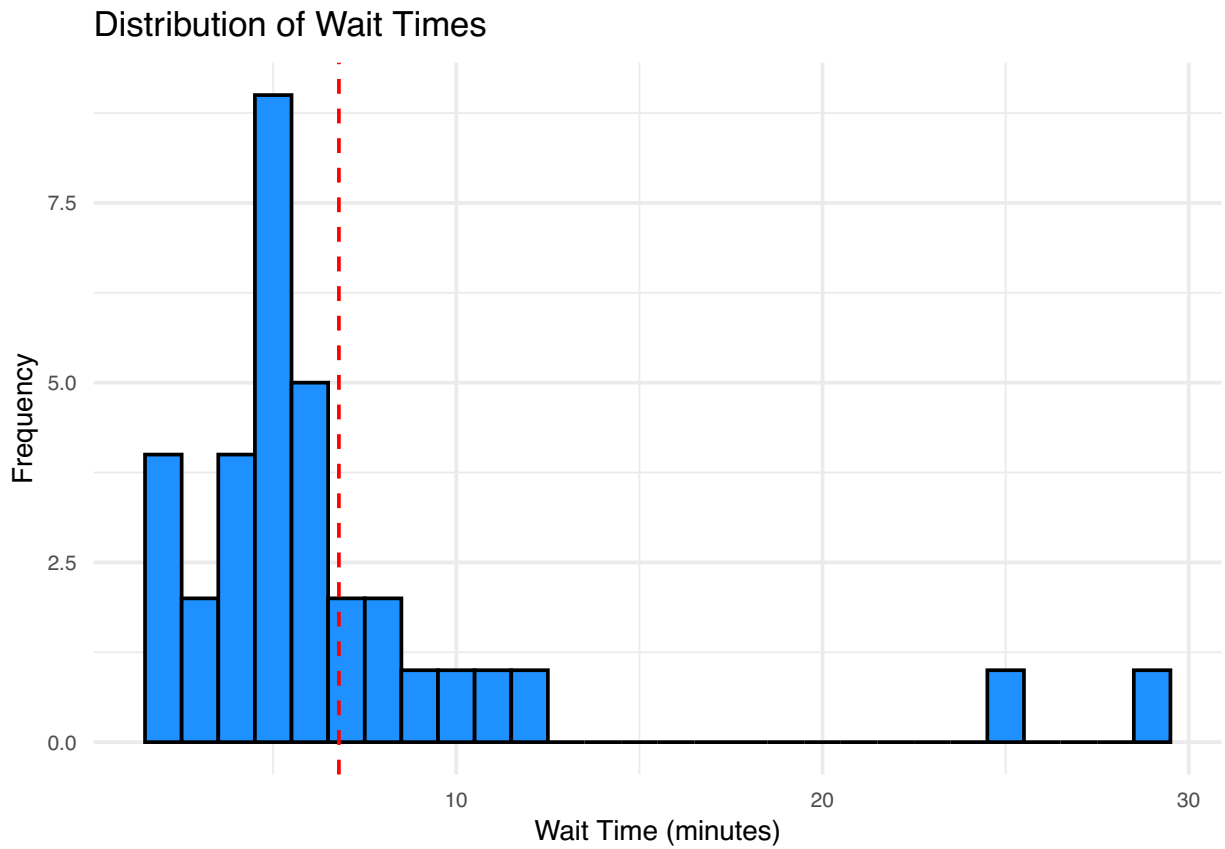
```r
# Load necessary libraries
library(tidyverse)
library(knitr)
library(kableExtra)

# Load data
cafeteria <- read_csv("clean_cafeteria_survey_2025.csv")

# Glimpse data
glimpse(cafeteria)
```

```
Rows: 35
Columns: 11
$ Wait_Time        <dbl> 4, 6, 5, 8, 2, 2, 9, 10, 4, 12, 5, 5, 7, 2, 6, 5, 5, ~
$ Satisfaction     <chr> NA, "Somewhat dissatisfied", "Somewhat dissatisfied",~
$ Satisfaction_num <dbl> NA, 2, 2, 5, 4, 4, 1, 4, NA, 4, 4, 5, 5, 4, 2, 4, 5, ~
$ Avoided_Bar      <chr> "Yes", "No", "No", "Yes", "Yes", "Yes", "Yes", "Yes",~
$ d_Avoided        <dbl> 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1,~
$ Payment_Method   <chr> "Mobile Phone", "Mobile Phone", "Mobile Phone", "Mobi~
$ Often_Canteen    <chr> "Every day", "Never", "Once a week", "Never", "Never"~
$ Fast_Track_Share <dbl> 50, NA, 0, NA, NA, NA, 72, NA, NA, 100, NA, NA, 100, ~
$ Wait_Fast_Track  <dbl> 5, NA, 0, NA, NA, NA, 2, NA, NA, 13, 0, 0, 7, NA, 9, ~
$ Wait_Reg_Track   <dbl> 12, NA, 0, NA, NA, NA, 11, 10, NA, NA, 15, 4, NA, NA,~
$ Nationality      <chr> "Portugal", "Germany", "Portugal", "Other", "Other", ~
```

Real Data Example: Estimating Wait Times



Mean Wait Time	SD	N	SE	LB	UB
6.79	5.7	34	0.98	4.81	8.78

How Confidence Intervals Work

A **confidence interval** gives us a range of plausible values for the true average wait time — guiding **better staffing and scheduling decisions**.

A confidence interval is generally built around a **point estimate** (like $\bar{Y}$) by adding and subtracting a **margin of error (ME)** to create a range:

$$(\hat{\theta} - \text{ME}, \hat{\theta} + \text{ME})$$

The **confidence level** tells us how often this method captures the true value if we repeat sampling.

Common confidence levels:

- 90% (narrower interval, more uncertainty)
- 95% (standard)
- 99% (wider interval, more conservative)

Confidence Intervals for the Mean (1/2)

Which formula we use depends on:

- Whether the population variance (σ^2) is known (usually it's not)
- The **distribution** of the population
- The **sample size** (n)

	σ^2 **is known**	σ^2 **is unknown**
Pop. Distribution: $Y \sim N(\mu, \sigma^2)$ **Sample size: Any n**	$\frac{\bar{Y}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$ CI: $\bar{Y} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\frac{\bar{Y}-\mu}{S'/\sqrt{n}} \sim t_{(n-1)}$ CI: $\bar{Y} \pm t_{\alpha/2}\frac{S'}{\sqrt{n}}$
Pop. Distribution: **Any with** $E[Y]=\mu$ **and** $V(Y)=\sigma^2$ **Sample size: Large n** ($n \geq 30$)	$\frac{\bar{Y}-\mu}{\sigma/\sqrt{n}} \dot{\sim} N(0,1)$ CI: $\bar{Y} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$	$\frac{\bar{Y}-\mu}{S'/\sqrt{n}} \dot{\sim} N(0,1)$ CI: $\bar{Y} \pm z_{\alpha/2}\frac{S'}{\sqrt{n}}$

Confidence Intervals for the Mean (2/2)

Usually:

If population distribution is Normal (any n):

$$\bar{Y} \pm t_{\alpha/2} \cdot \frac{S'}{\sqrt{n}}$$

If population distribution is unknown and $n \geq 30$ (by CLT):

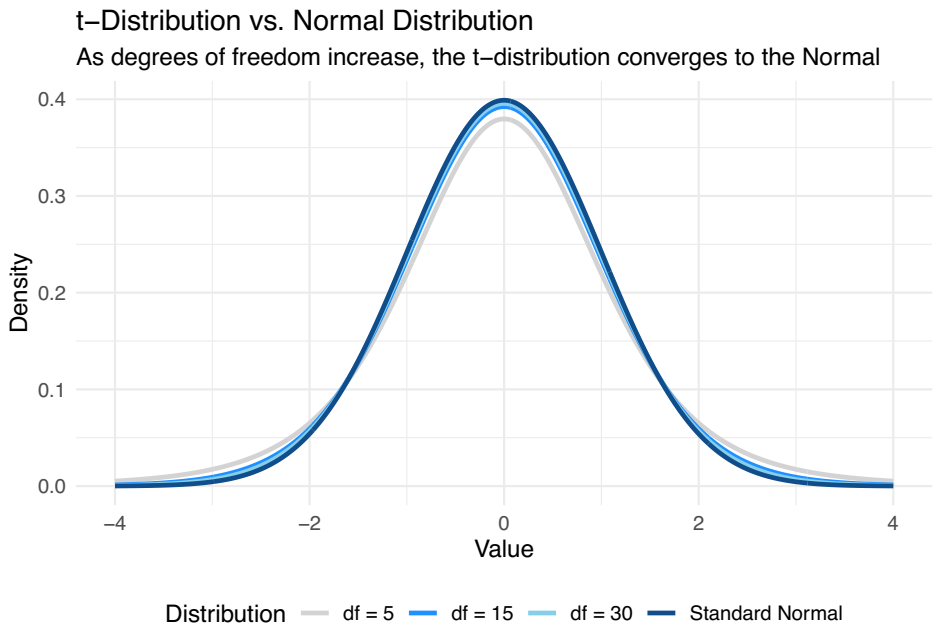
$$\bar{Y} \pm z_{\alpha/2} \cdot \frac{S'}{\sqrt{n}}$$

Where:

- $\bar{Y}$ = Sample mean (e.g., mean wait time)
- S' = Sample standard deviation (e.g., standard deviation of wait time)
- n = Sample size (e.g., number of respondents)
- $z_{\alpha/2}$ or $t_{\alpha/2}$ = Critical value from standard or t distribution (default: 1.96)

Visualizing the t-Distribution

- The **t-distribution** is wider than the **Normal**, especially for small n, reflecting **greater uncertainty**.
- As n grows, the t-distribution **converges to the Normal**.
- That's why `t.test()` in R always uses the **t-distribution** – it's valid for small and large samples alike.



What Does "95% Confidence" Actually Mean?

The Big Idea:

- A **confidence interval** is random — the **method** is what's reliable.
- The **confidence level** is the long-run success rate of the **method**:
  - If we repeated our cafeteria wait time study 100 times, about **95 of the resulting intervals** would contain the **true average wait time**.

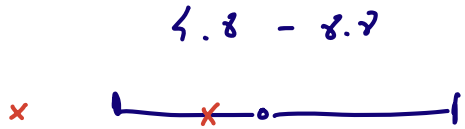
What this does *not* mean:

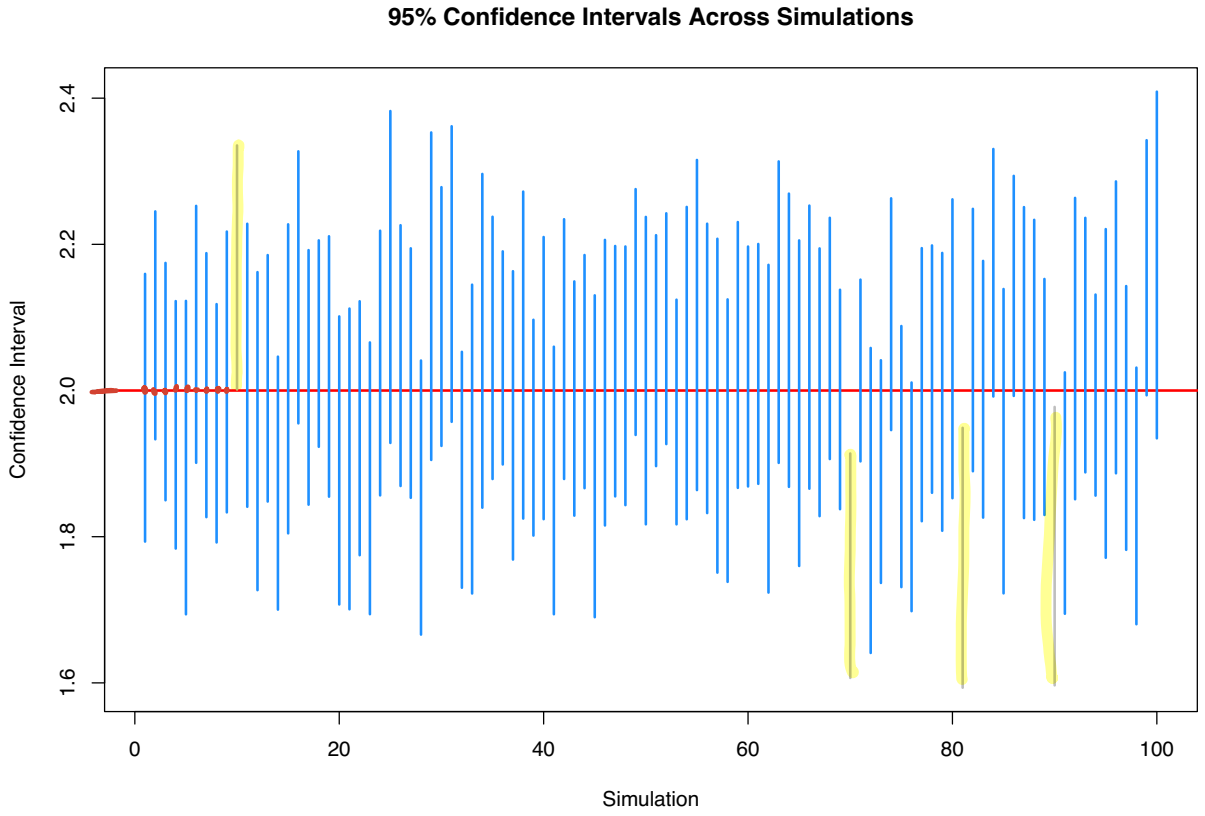
- It does *not* mean there's a 95% chance the true value is in *this* interval
- Once you have your interval, it either contains μ or it doesn't — no probabilities apply

Simulation – Intervals Across 100 Samples

Let's **simulate 100 samples**, and for each we compute a 95% confidence interval for the mean wait time in the cafeteria line.

- **Blue intervals** contain the true value
- **Gray intervals** miss it
- **Red line** shows the true mean wait time





Manual CI Calculation

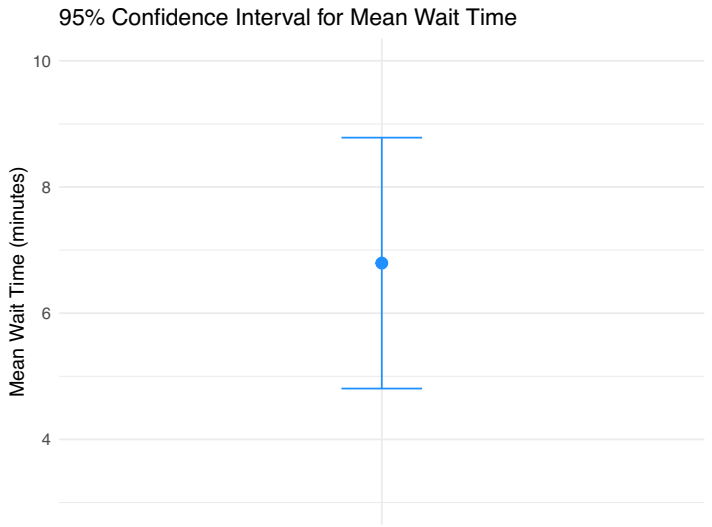
Now we are ready to demystify what's behind `t.test()`. Let's compute the CI manually using the formula:

$$(\bar{Y} \pm t_{\alpha/2} \cdot \frac{S}{\sqrt{n}})$$

$6.79 \pm 2.03 \times \frac{5.7}{\sqrt{34}} \rightarrow 6.79 \pm 2.03 \times 0.98$

Mean Wait Time	SD	N	t*	SE	LB	UB
6.79	5.7	34	2.03	0.98	4.81	8.78

$(4.81 ; 8.78)$



Recap: Why Confidence Intervals Matter

A **point estimate** gives a single best guess but tells us nothing about how **uncertain or precise** that guess is.

A **confidence interval** shows a **range** of plausible values, accounting for **sampling variability** and **sample size**.

- We estimate the average **wait time in line** is between **5.8 and 8.8 minutes**
- This gives us a **confidence-based window**, not just a guess.

But what if we want to **go further**?

- Did the **average wait time improve from 2024 to 2025**?

That's where **hypothesis testing** comes in — it helps us make data-backed decisions.

4. Hypothesis Testing

Binding 2024 and 2025 Survey Data

```
# Load data
cafeteria.2024 <- read_csv("clean_cafeteria_survey_2024.csv")
cafeteria.2025 <- read_csv("clean_cafeteria_survey_2025.csv")

# Select columns of interest
cafeteria.2024 <- cafeteria.2024 |> select(Wait_Time, Payment_Method, Satisfaction, Nationality) |> mutate(Year = 2024)
cafeteria.2025 <- cafeteria.2025 |> select(Wait_Time, Payment_Method, Satisfaction, Nationality) |> mutate(Year = 2025)

# Bind data
cafeteria.all <- bind_rows(cafeteria.2024, cafeteria.2025)

# Average Wait Times by Year
cafeteria.all |> group_by(Year) |> summarize(
    mean_wait_time = mean(Wait_Time, na.rm = TRUE),
    sd_wait_time = sd(Wait_Time, na.rm = TRUE),
    n = sum(!is.na(Wait_Time)))
```

```
# A tibble: 2 x 4
   Year mean_wait_time sd_wait_time     n
  <dbl>          <dbl>        <dbl> <int>
1  2024           5.33         4.76    42
2  2025           6.79         5.70    34
```

From Uncertainty to Decisions: Testing What Matters

We've estimated students' **average wait times** in the cafeteria line:

- 2024: ~5.3 minutes
- 2025: ~6.8 minutes

But, are those observed **differences** in wait times **statistically meaningful** or just random variation?

The cafeteria manager wants to know:

- Did average wait times really increase from 2024 to 2025?
- Are differences across years large enough to matter, or just noise?

To answer confidently, we turn to **hypothesis testing.**

Why Hypothesis Testing?

Confidence intervals give us a **range**, but not a **verdict**.

To support decisions, hypothesis tests allow us to:

- Formally test **claims about group differences**
- Quantify **evidence** against the status quo (null)
- Make **confident, data-driven recommendations**

This is how we distinguish **signal** from **noise** — and avoid jumping to conclusions.

$\bar{y}_{25} = 6.8$ $\quad \Delta\ 1.5$

$\bar{y}_{24} = 5.3$

$H_0: \mu = 0$ $\quad H_a: \mu \neq 0$ $\quad \bar{Y}$

$H_0: \sigma^2 \geq 5$ $\quad H_a: \sigma^2 < 5$ $\quad S^{*2}$

$H_0: \mu_1 = \mu_2$ $\quad H_a: \mu_1 \neq \mu_2$ $\quad \bar{Y}_1 - \bar{Y}_2$

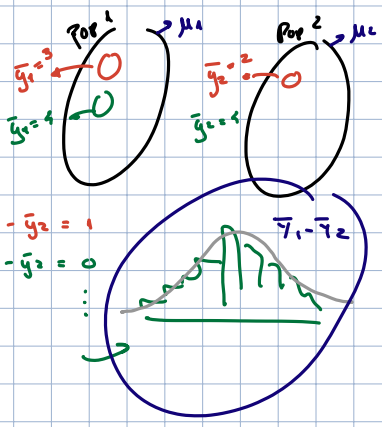
2 Hypotheses

Null: H_0: no effect; no change; no difference; no relationship

$= \geq \leq$ $\quad \mu_{25} = \mu_{24}$ $\quad \mu_{25} - \mu_{24} = 0$

Alternative: H_1 or H_a : $\mu_{25} \neq \mu_{24}$

$\neq < >$

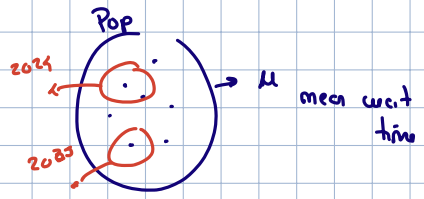


2 Decision

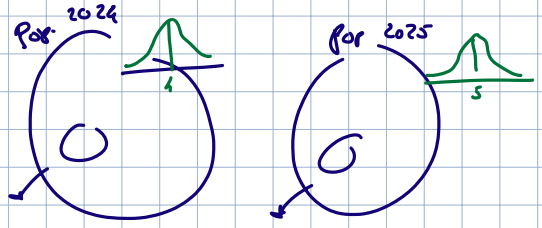
Reject: the evidence supports the alternative hypo.

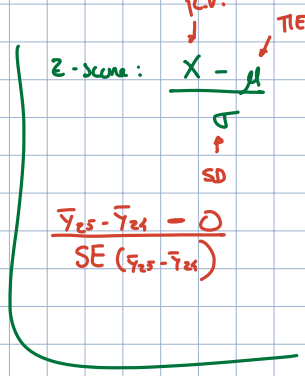
Fail to reject: " does not " "

Decide: Are these from same dist?



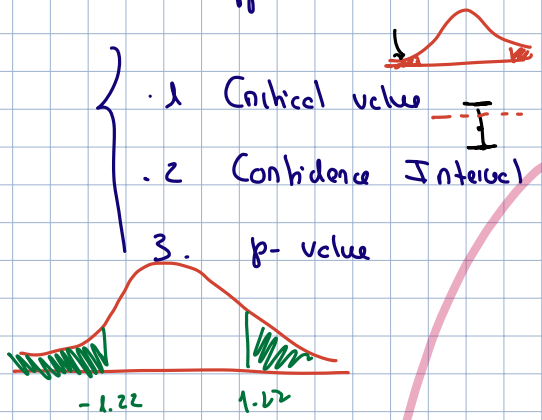
or from different dist?





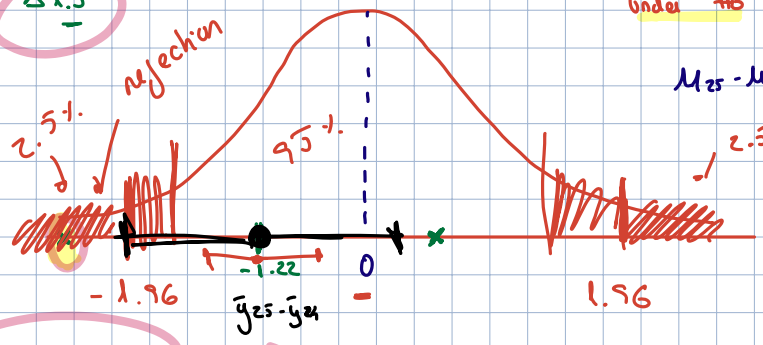
1 test – 3 equivalent approaches

1. Critical value
2. Confidence Interval
3. p-value



Sampling Dist. $\bar{Y}_{25} - \bar{Y}_{24}$ under H_0

$\mu_{25} - \mu_{24} = 0$



$\text{test stat} = \frac{1.5 - 0}{SE} = 1.22$

CI approach: If interval contains null value ($\mu_{25} - \mu_{24} = 0$)

If interval does not contain null value → reject

$H_0: \mu = 25$

(20, 30)

What Is a Hypothesis?

A statistical hypothesis is a **testable claim** about:

- A single **population parameter** (e.g., an average wait time), or
- A **comparison of parameters** (e.g., difference in average wait times across years)

Ex.: Did the average cafeteria wait time change from 2024 to 2025?

We formulate two hypotheses:

- **Null hypothesis (H_0)**: the default "no difference" or "no effect" claim; this is what we test directly.
  - $H_0 : \mu_{2024} = \mu_{2025}$ (no difference)
- **Alternative hypothesis (H_a)**: the competing claim we seek evidence to support.
  - $H_a : \mu_{2024} \neq \mu_{2025}$ (real difference)

The 5-Step Hypothesis Testing Procedure

1. State the **null** (H_0) and **alternative** (H_a) hypotheses about a **parameter** or **comparison** of **parameters**;
2. Choose the **significance level**, α;
3. Use sample data to compute a **test statistic** that measures the **evidence against** H_0;
4. Compute the **p-value**;
5. If the *p-value* $< \alpha$, **reject** H_0 (the evidence supports H_a); if the *p-value* $> \alpha$ **fail to reject** H_0 (there is insufficient evidence against H_0).

This process helps us translate sample results into decisions.

Note: Failing to reject H_0 does **not** imply that the null is true.

From Sample to Test Statistic

A **test statistic** tells us how far our observed value is from what we'd expect under the **null hypothesis**, measured in **standard errors**.

General formula:

$$\text{Test Statistic} = \frac{\text{Sample Estimate} - \text{Null Value}}{\text{Standard Error}}$$

–> You've seen this before: it's the **same idea** as a **z-score** — standardizing a value to see how unusual it is.

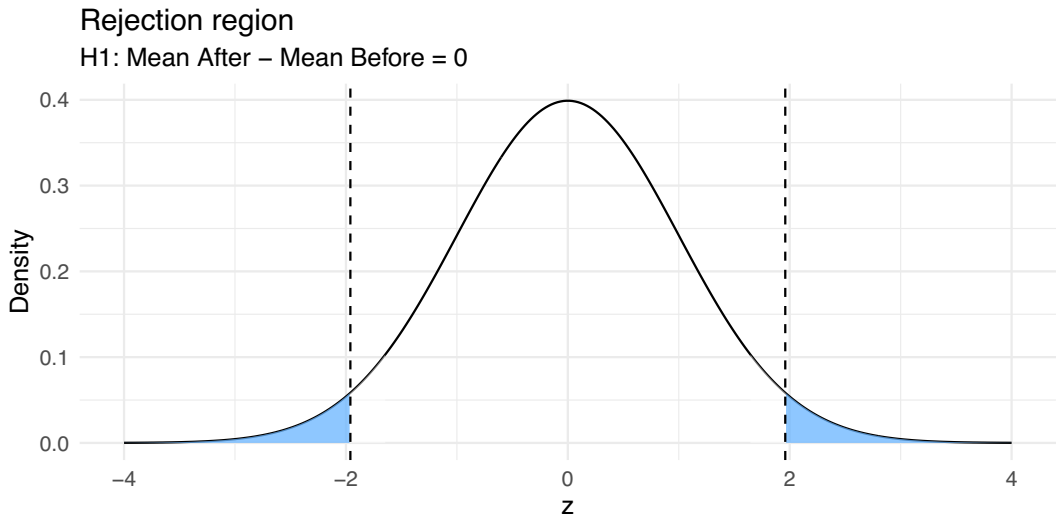
- The **larger** (in absolute value), the stronger the evidence **against** H_0
- The **sign** indicates the **direction** of the difference

Decision Rule: Critical Values & Rejection Regions

- The **rejection region** defines the values of the test statistic that would lead us to reject H_0.
- The **critical value(s)** is/are the cutoff point(s) for what's considered "too extreme" under H_0.
  - It depends on the significance level (α) and test type (1-tailed or 2-tailed)

$(1-\alpha)$ ← Conf. level: 90%, 95%, 99%

Sig level → α: 10%, 5%, 1%



The curve above shows the **sampling distribution** of the test statistic **under** H_0 – the values we'd expect by chance if the null were true.

Comparing Two Means in R

Ex.: Did average cafeteria wait times change from 2024 to 2025?

- $H_0 : \mu_{2024} = \mu_{2025}$ (no difference)
- $H_a : \mu_{2024} \neq \mu_{2025}$ (real difference)

Year	Mean Wait Time	SD	N	SE
2024	5.333333	4.760952	42	0.7346309
2025	6.794118	5.698375	34	0.9772633

```
# Perform t-test
cafeteria.all |> t_test(Wait_Time ~ Year, var.equal = TRUE,
                         paired = FALSE, alternative = "two.sided")
```

```
# A tibble: 1 x 8
  .y.       group1 group2    n1    n2 statistic    df     p
* <chr>     <chr>  <chr>  <int> <int>     <dbl> <dbl> <dbl>
1 Wait_Time 2024   2025      42    34     -1.22    74 0.227
```

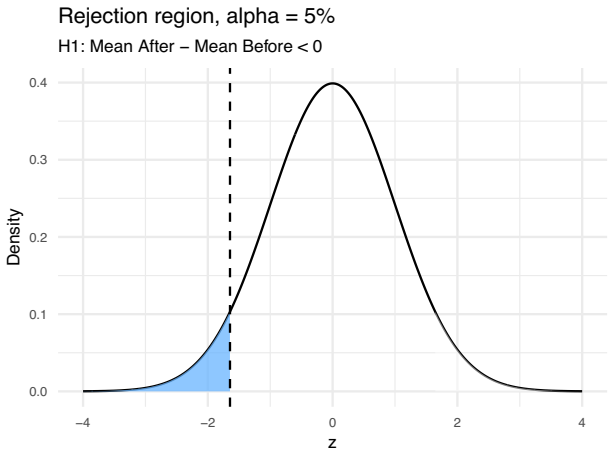
One Tailed Tests

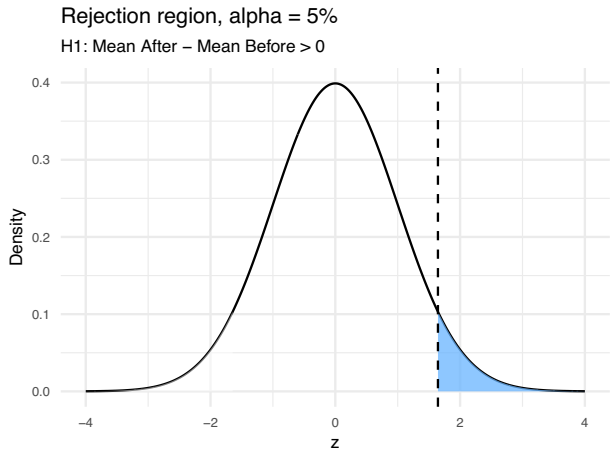
```
# One-tailed test
cafeteria.all |> t_test(Wait_Time ~ Year, var.equal = TRUE,
                        paired = FALSE, alternative = "greater")
```

```
# A tibble: 1 x 8
  .y.       group1 group2    n1    n2 statistic    df     p
* <chr>     <chr>  <chr>  <int> <int>     <dbl> <dbl> <dbl>
1 Wait_Time 2024   2025      42    34     -1.22    74 0.886
```

```
# One-tailed test
cafeteria.all |> t_test(Wait_Time ~ Year, var.equal = TRUE,
                        paired = FALSE, alternative = "less")
```

```
# A tibble: 1 x 8
  .y.       group1 group2    n1    n2 statistic    df     p
* <chr>     <chr>  <chr>  <int> <int>     <dbl> <dbl> <dbl>
1 Wait_Time 2024   2025      42    34     -1.22    74 0.114
```





Understanding p-values

The **p-value** is the probability — assuming the null hypothesis (H_0) is true — of observing a test statistic **as extreme or more extreme** than the one we got.

- A **small p-value** means our result would be **unlikely** if H_0 were true $\rightarrow$ evidence **against** H_0.
- A **large p-value** suggests the data is **consistent** with $H_0 \rightarrow$ we **fail to reject**.

Decision Rule:

- **Reject** H_0 if: p-value α (e.g., 0.05)
- This tells us whether the result is **statistically significant**.

Ethical Reminder: choosing α **after** seeing the p-value (to force a result) is **bad practice**!

Hypothesis testing

1. Formulate hypotheses

2 Hypotheses { Null : H_0 (=, ≥, ≤)
Alternative : H_a (≠, <, >)

About 1 parameter : $H_0: \mu \leq 5$ vs $H_a: \mu > 5$

$H_0: p \leq 0$ vs $H_a: p > 0$

2 parameters : $H_0: \mu_{25} = \mu_{24} \rightarrow \mu_{25} - \mu_{24} = 0$

$H_a: \mu_{25} \neq \mu_{24}$

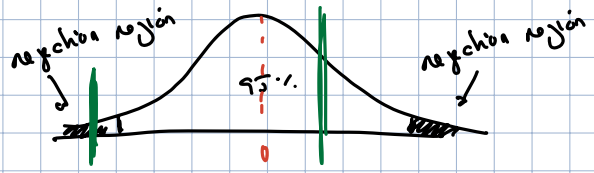
2. Choose α (sig. level)

1 %
5 % → default
10 %.

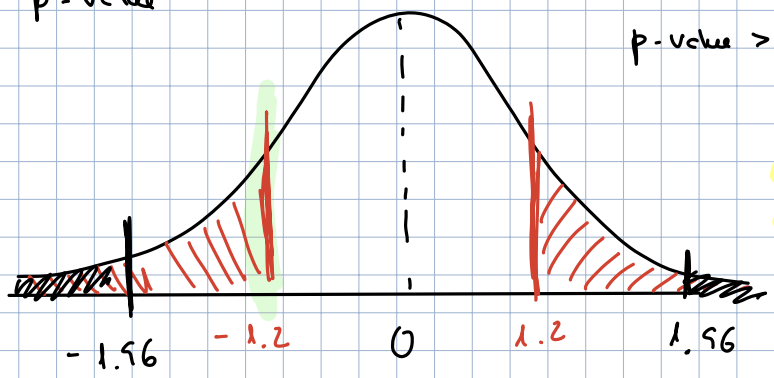
3. Compute a test-statistic

test stat (z or t score) = $\dfrac{\text{estimate} - \text{null value}}{\text{SE estimate}}$

4.1 Critical region



4.2 p-value



p-value > α → Fail to reject

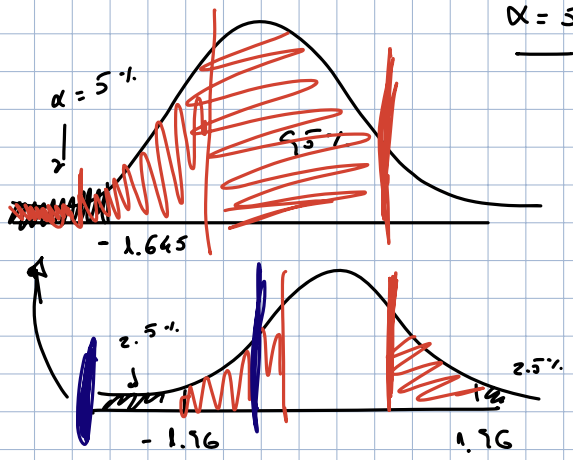
p-value = 0.000001 < 0.05 ✓
0.01 ✓

If p-value < α → Reject H_0

H_0: D.H ≥ 0

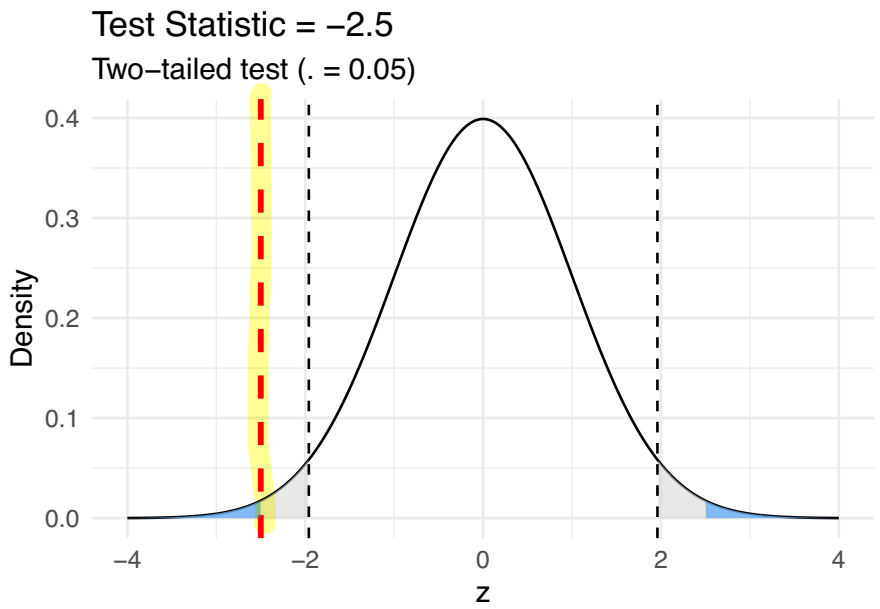
H_1: D.H < 0

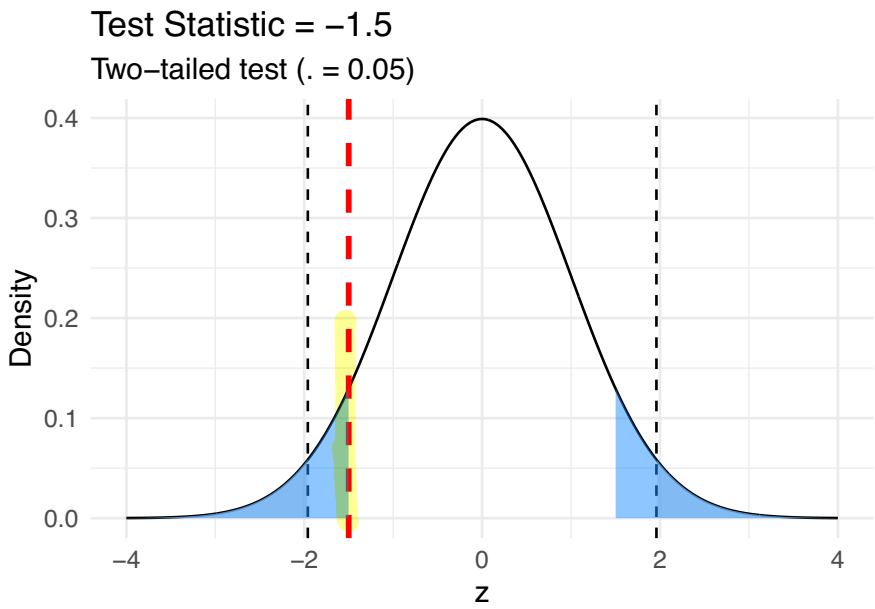
$\alpha = 5\%$

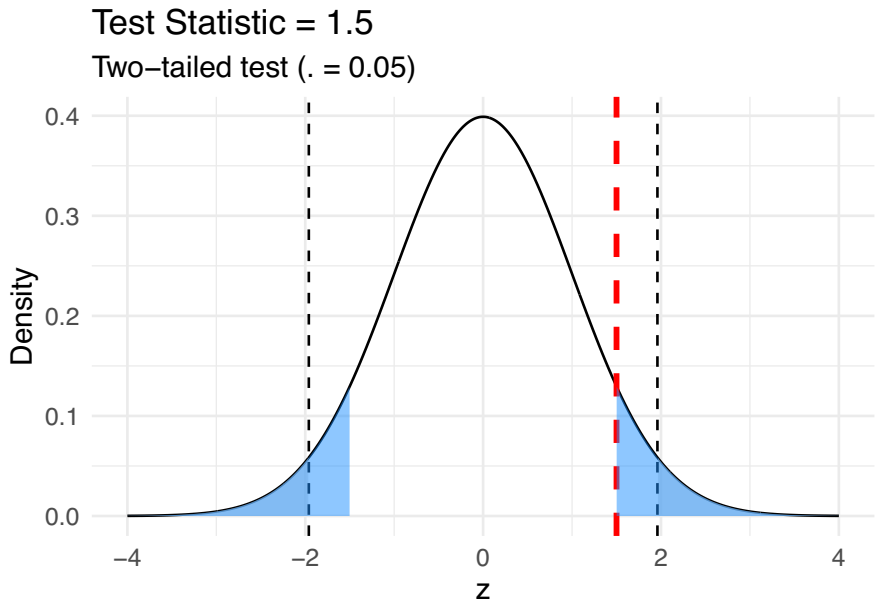


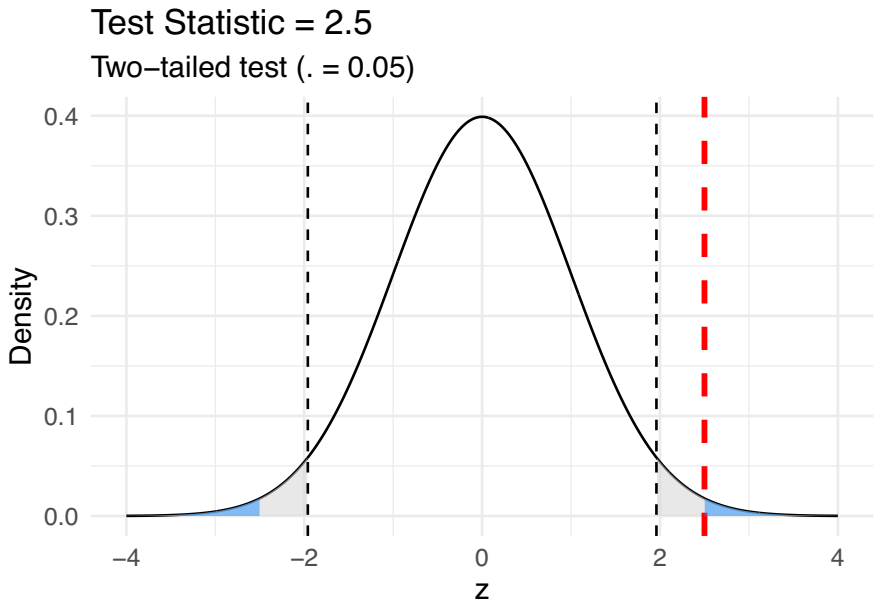
Decision : p-value $< \alpha$ → Reject H_0

One-Sided and Two-Sided p-values









When Is Hypothesis Testing Valid?

To trust our p-values and conclusions, we rely on these assumptions:

1. **Random sample** from the population
2. **Independence**: responses aren't linked
3. **Normality**: each group's data is (roughly) normal
    - *Important for small samples*
4. The variances in the different groups should be similar (**homogeneity of variances**).

Note:

- For larger n, the **Central Limit Theorem** makes normality less critical
- If assumptions break, consider **non-parametric** alternatives (e.g. Wilcoxon, permutation tests)

Confidence Intervals & Hypothesis Testing

A **confidence interval** shows the range of parameter values consistent with the data.

- If the **null value** lies **inside** the interval $\rightarrow$ we **fail to reject** H_0
- If the null value lies **outside** the interval $\rightarrow$ we **reject** H_0

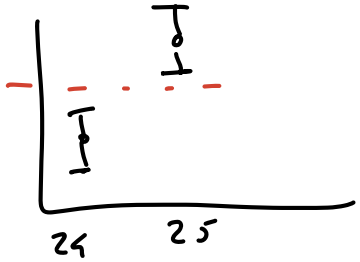
This provides a useful **visual check** for hypothesis test conclusions.

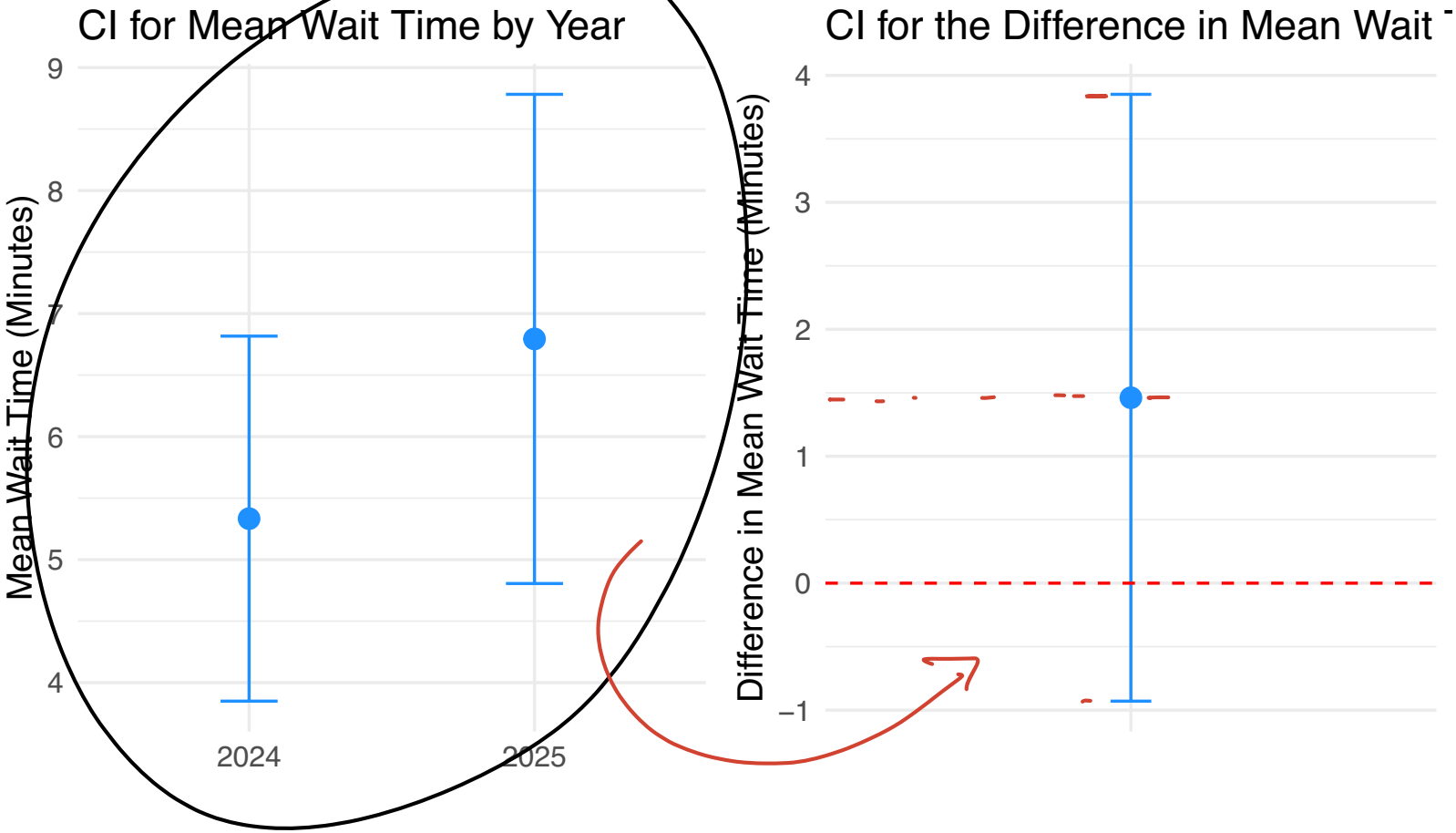
```
# Comparing two means: Wait time before vs after
t.test(cafeteria.all$Wait_Time ~ cafeteria.all$Year,
       var.equal = TRUE)
```

```
    Two Sample t-test

data:  cafeteria.all$Wait_Time by cafeteria.all$Year
t = -1.2177, df = 74, p-value = 0.2272 > 0.05 → Fail to reject
alternative hypothesis: true difference in means between group 2024 and group 2025 is not e
95 percent confidence interval:
 -3.8510517  0.9294831
sample estimates:
mean in group 2024 mean in group 2025
          5.333333           6.794118
```

CI for Difference in Means





Comparing Multiple Means: Analysis of Variance (ANOVA)

ANOVA is a generalization of the **t-test**, suitable when we have one **categorical variable** with ≥ 3 levels (groups) and one **continuous outcome**.

Hypotheses:

- H_0: All group means are equal
- H_a: At least one group mean is different

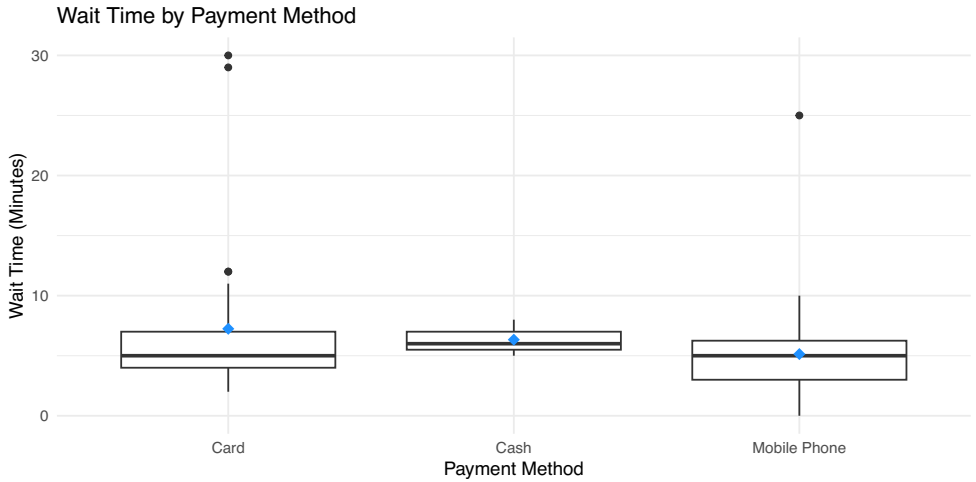
Key Assumptions:

- **Independence** of observations
- **Normality** within each group
- **Homogeneity of variances** (equal group variances)

If the ANOVA test is significant, it tells us **some difference exists** — but **not which groups differ**. For that, we follow up with **post-hoc tests**.

ANOVA in R (real survey data)

Goal: Test whether **average wait times** differ by **payment method**.




```r
# Run ANOVA to compare wait times across payment methods
anova_result <- aov(Wait_Time ~ Payment_Method, data = cafeteria.all)

# Display ANOVA summary
summary(anova_result)
```

```
               Df Sum Sq Mean Sq F value Pr(>F)
Payment_Method  2   77.8   38.91   1.447  0.242
Residuals      73 1963.2   26.89
2 observations deleted due to missingness
```

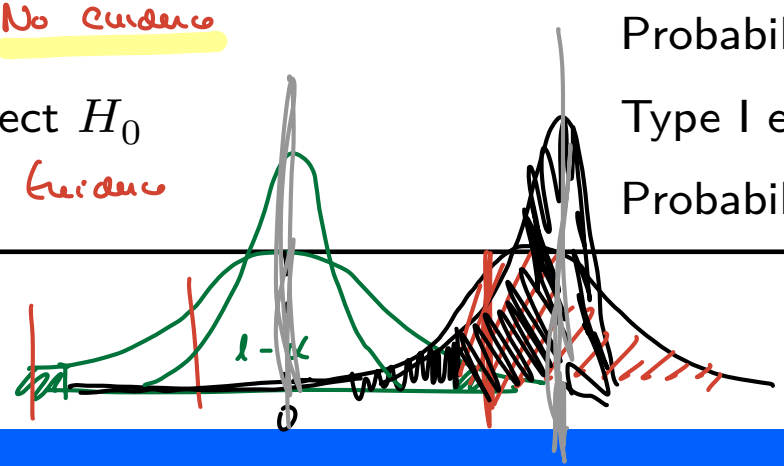
Type I and type II errors

When we make decisions using hypothesis tests, **errors can occur**:

- We might **reject a true null hypothesis** (false alarm)
- Or **fail to reject a false null hypothesis** (miss a real effect)

Table 4: Decisions on Null Hypothesis and States of Nature

Decisions on Null Hypothesis	**States of Nature**	
	Null Hypothesis Is True	**Null Hypothesis Is False**
Fail to reject H_0	Correct decision	Type II error
	Probability $= 1 - \alpha$	Probability $= \beta$
Reject H_0	Type I error	Correct decision
	Probability $= \alpha$	Probability $= 1 - \beta$



Practical vs. Statistical Significance

A **statistically significant** difference means the data provides enough evidence to reject the null — **but** that doesn't mean the difference is **important**.

- Always ask: Is this difference **large enough** to matter for operations?
- If average wait time in 2025 is 6.8 minutes and in 2024 is 5.3 minutes is the 1.5 minutes **meaningful**?
- If so, this could justify:
  - Adding **extra staff** during peak times
  - **Adjusting schedules** around class breaks
  - Investment in better **queue management systems**

Statistical tools help us detect **signal**, but **management judgment** tells us whether the signal matters.

Recap — From Estimates to Evidence

1. **Point Estimation**
   - Estimating quantities like **average wait times**
2. **Sampling Distributions**
   - Understanding variation across samples
   - Concepts like **bias**, **standard error**, and the **Central Limit Theorem**
3. **Confidence Intervals**
   - Providing a **range of plausible values**
   - Quantifying **uncertainty** in a clear and interpretable way
4. **Hypothesis Testing**
   - Evaluating claims and detecting **statistical signals**
   - Comparing groups and testing effects

Next Steps: From Averages to Relationships

So far, we've focused on **comparing group means** like wait times across years.

What if we want to explore the underlying factors shaping wait times?

- How do **multiple factors** together influence waiting?
- Can we isolate one effect while **controlling for others**?

Next: Regression Analysis

- A tool to uncover how **outcomes respond to inputs**
- Lets us model and explain **relationships**
- Works with **multiple variables at once**

Instead of asking: "Are these means different?"
We'll ask: ***"How does Y change when X changes?"***

Recommended Readings

- Wooldridge, *Introductory Econometrics*, Math Refresher B: Fundamentals of Probability
  - C-1 Populations, Parameters, Random Sampling
  - C-2 Finite Sample Properties of Estimators
  - C-3 Asymptotic or Large Sample Properties of Estimators
  - C-5 Interval Estimation and Confidence Intervals
  - C-6 Hypothesis Testing